

Rahul Kumar

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EDUCATION

• Vellore Institute of Technology Bhopal

Bachelor of Technology in Computer Science and Engineering

- GPA: 8.45/10

Bhopal, India

Sep. 2024 – Present

SKILLS

- **Languages & Markup:** HTML5, CSS3, JavaScript (ES6+), TypeScript
- **Frameworks & Libraries:** React.js, Tailwind CSS, Framer Motion, React Hook Form, Recharts, Zod, Socket.IO
- **Backend (Growing):** Node.js, Express.js, REST APIs, PostgreSQL (Supabase), MongoDB, JWT Auth
- **Tools & Platforms:** Git, GitHub, Vite, Vercel, Render, VS Code, esbuild
- **Concepts:** Component Architecture, State Management, Web Performance (FCP/LCP/CLS), Responsive Design, WebSockets, RBAC
- **Soft Skills:** Problem Solving, Adaptability, Effective Communication, Team Collaboration, Time Management

EXPERIENCE

Front End Software Engineer Intern

GNB Motors Private Limited

May 2026 – Present

React.js, JavaScript, TypeScript, Vite, CSS, Git

- Architected a full-stack **Role-Based Access Control (RBAC)** system from scratch — backend middleware enforcing hierarchical permissions and dynamic frontend rendering that conditionally gates features by role, eliminating manual access checks across the codebase.
- Designed and shipped a **custom geofencing system** end-to-end: interactive map UI (AddZoneDrawer) for visually drawing zones, real-time backend engine monitoring vehicle positions against boundaries, and unit tests catching edge-case alert trigger bugs.
- Engineered an iOS-style sliding pill filter on the overview dashboard with optimized state management, enabling seamless day-change data filtering without hard page refreshes.
- Redesigned user onboarding into a 5-step guided flow with a 3-layer Chrome extension detection mechanism and visual FleetEdge setup guides, reducing new-user friction at first run.
- Resolved 4+ critical UI defects including action dropdown clipping, persistent sidebar state corruption, and data-table scroll lock — each traced, root-caused, and fixed without regressions.

PROJECTS

EventFlow (QLess) | *React.js, TypeScript, Node.js, Supabase (PostgreSQL), Socket.IO, Razorpay* | [Live](#)

- Built a **multi-tenant SaaS** event management platform for university clubs featuring QR-based ticketing, real-time gate scanning via WebSockets, Razorpay payment integration, and an analytics dashboard (Recharts + Socket.IO feeds).
- Deployed for Jhalak '26 — processed **350+ student entries with zero scan failures**, enforcing atomic single-scan QR validation at the database level to prevent duplicate entries.
- Implemented React Hook Form + Zod validation, html5-qrcode in-browser scanner, stateless JWT volunteer sessions, and Google Sheets bulk import for event organizers.

IPL 2026 Stats Dashboard | *React.js, Node.js, MongoDB, Socket.IO, Puppeteer, Gemini AI* | [Live](#)

- Eliminated cloud crashes from 200MB+ Puppeteer instances by replacing headless scraping with a **3-source fallback pipeline** (ESPN Cricinfo → Cricbuzz → DOM regex), cutting memory by **80%+** on a 40-second live polling interval.
- Tracked 10 real-time match metrics (CRR, RRR, win probability, normalized overs) with a 20-minute freeze-state window handling rain delays, innings switches, and double-header shifts.
- Isolated heavy stat imports (Orange Cap, Purple Cap, NRR recalculation) into a separate 12-hour cron service, preventing interference with the live match data layer.

Black Diamond Motors | *Next.js, TypeScript, Tailwind CSS, Framer Motion* | [Live](#)

- Rebuilt corporate vehicle site on Next.js App Router achieving **FCP of 302ms** — **4× faster** than the original site's 1,224ms under identical network conditions (verified via Chrome trace files).
- Delivered **zero CLS (0.0000)** vs. 0.0099 on the original by structuring components to eliminate cumulative layout shift; implemented Framer Motion cinematic transitions across all routes.

ADDITIONAL INFORMATION

- **Languages:** English (Professional Working Proficiency), Hindi (Fluent)
- **Extracurricular:** Core Member, Gangabhumii Club Tech Team, VIT Bhopal — managed technical event systems, website maintenance, and campus event coordination.